The Power of Structural Rules: Proof-Size Lower Bounds for Linear Logics

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28 October, 2026



Outline

1. Motivation
2. Intuitionistic Substructural Logics
3. Classical substrcutural logics

"When I started with logic, I considered propositional logic absolutely uninteresting. I wondered why some logicians pay so much attention to the laws and rules of propositional logic when there is a simple procedure to test whether a propositional formula is a tautology. This view conformed with the widespread opinion that mathematics starts with infinity because finite problems are trivial. We can indeed check a tautology simply by testing all truth assignments to its variables, but when we consider the complexity of testing, things become interesting (or if we consider non-classical logics, which I will not do in this paper)."

P. Pudlák: Proofs, computational complexity, and games, Proceedings ECM 24

Aim

- Decision/Computational Complexity: Given a formula A, is it valid? Algorithms (resources: time, space, ...)
- Proof Complexity: Given a tautology A and a proof system P, what is the shortest proof of A in P?
  Proofs (resources: length, depth, ...)

A useful analogy:

Area	**Question**
Computability theory	Does an algorithm exist?
Computational complexity	How efficient is the algorithm?
Proof theory	Does a proof exist?
Proof complexity	How efficient (short) can the proof be?

Abstract Proof Systems

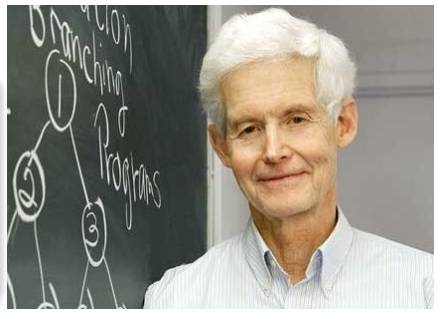
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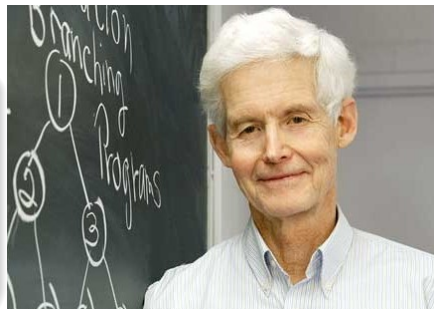
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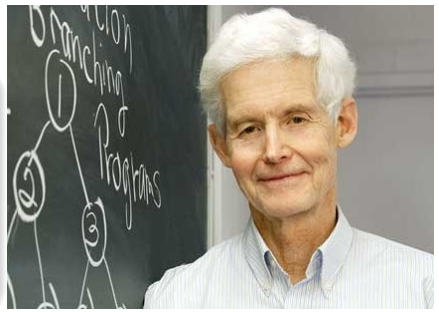
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Examples: The usual Hilbert-style and Gentzen-style proof systems.

Connection to complexity theory

p-bounded pps: all tautologies have polynomial size proofs[1].

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- TAUT: coNP-complete. (Because SAT is NP-complete.)
- If $\exists n \; |y| \leqslant |\varphi|^n$, we get an NP set. □

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Cook's program

Conjecture: $\mathrm{NP} \neq \mathrm{coNP}$ (which implies $\mathrm{P} \neq \mathrm{NP}$)

Thus, we expect that there is no p-bounded pps, i.e., all proof systems have *hard theorems*.

Cook's Program: Separate NP from coNP

Show *super-polynomial* lower bounds to the lengths of proofs in **all** propositional proof systems for Classical Propositional Logic, CL.

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A sequence $\{\varphi_n\}_{n\in\mathbb{N}}$ of tautologies such that for any $n \in \mathbb{N}$, any P-proof of φ_n is super-polynomially long in $|\varphi_n|$.

Why non-classical logics?

Understanding proof complexity in non-classical logics matters because of:

- their various applications
- their power in expressibility
- their essential role in computer science
- the impact that lower bounds on lengths of proofs have on the performance of proof-search algorithms.

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- For many non-classical logics the analogue of Cook's program corresponds to NP vs PSPACE. (Non-classical logics are often more complex than CL (e.g. PSPACE-complete).)

Known Results

Classical realm: Weaker systems

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Non-classical realm: Change CL (e.g. forbid $\neg\neg A \to A$ or add modalities). Corresponds to NP vs PSPACE, which is supposedly more tractable than NP vs coNP. Confirmation to this intuition:

Theorem (Hrubeš '07, Jeřábek '09)

There are exponential lower bounds for intuitionistic and modal Frege systems.

Unfortunately, the theorem does not extend to CL.

Substructural systems

Motivation: Analogue of Cook's program

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Is the above theorem trivial?!

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Frege systems are the most natural calculi for propositional logic.

Frege system P: a finite set of rules, where a rule is an expression

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A **P**-proof of φ from a set of assumptions X, denoted by $X \vdash_{\mathbf{P}} \varphi$, is

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The formulas φ_i are called lines of the proof π, denoted by $\ell(\pi)$.

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- What do commas mean in a sequent calculus lacking (left) weakening or contraction or both?
- Introduce a new logical connective called fusion $*$.

Substructural calculi

Language: $\{0, 1, \wedge, \vee, *, \rightarrow\}$. Capital Greek letters: multisets of formulas.

$$\varphi \Rightarrow \varphi \qquad \Rightarrow 1 \qquad 0 \Rightarrow$$

$$\frac{\Gamma \Rightarrow \Delta}{\Gamma, 1 \Rightarrow \Delta} \qquad \frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow 0, \Delta}$$

$$\frac{\Gamma, \varphi \Rightarrow \Delta}{\Gamma, \varphi \wedge \psi \Rightarrow \Delta} \qquad \frac{\Gamma, \psi \Rightarrow \Delta}{\Gamma, \varphi \wedge \psi \Rightarrow \Delta} \qquad \frac{\Gamma \Rightarrow \varphi, \Delta \qquad \Gamma \Rightarrow \psi, \Delta}{\Gamma \Rightarrow \varphi \wedge \psi, \Delta}$$

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$$\frac{\Gamma, \varphi, \psi \Rightarrow \Delta}{\Gamma, \varphi * \psi \Rightarrow \Delta} \qquad \frac{\Gamma \Rightarrow \varphi, \Delta \qquad \Sigma \Rightarrow \psi, \Lambda}{\Gamma, \Sigma \Rightarrow \varphi * \psi, \Delta, \Lambda}$$

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Contraction rules:

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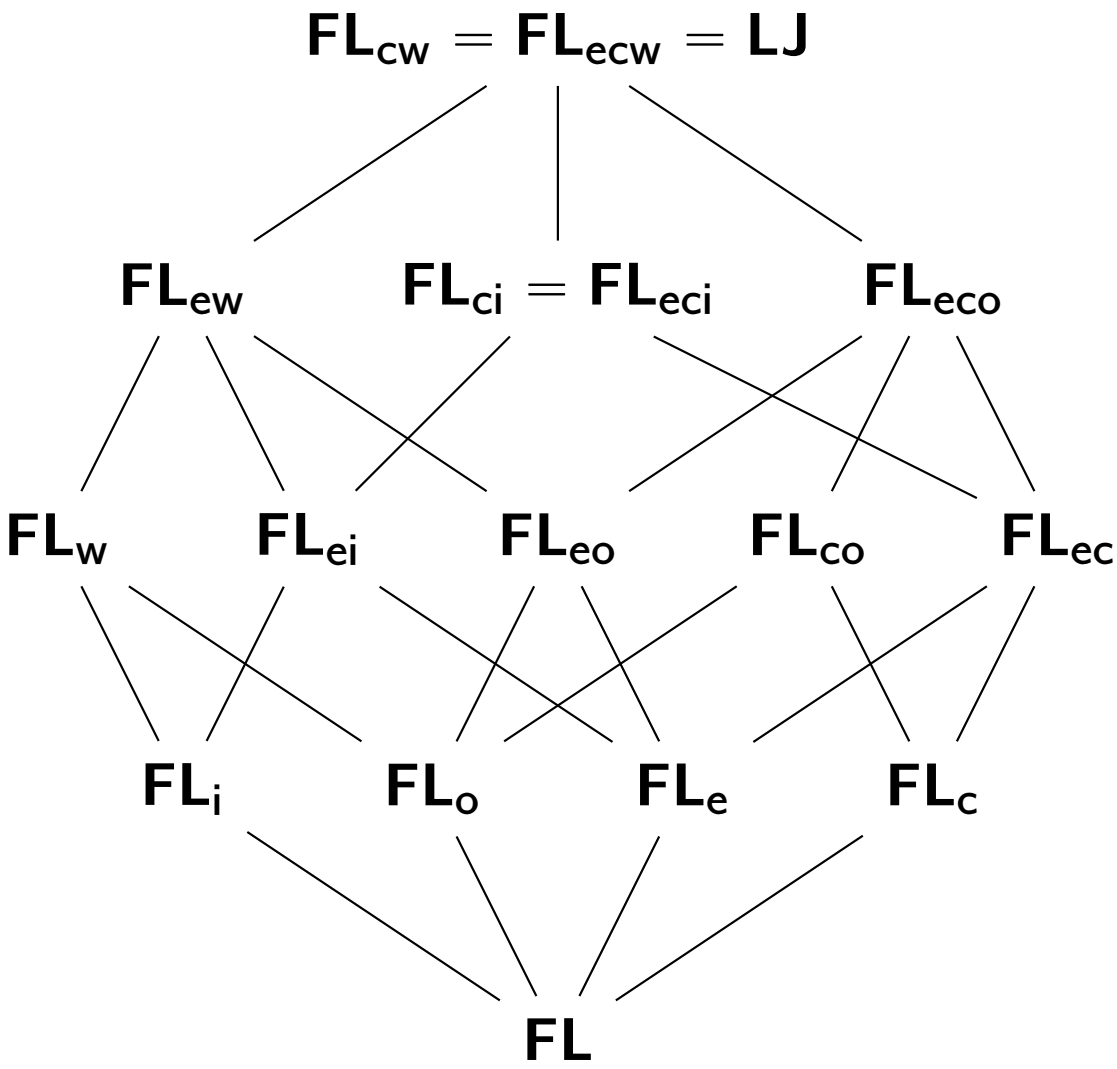
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- $\mathbf{FL_{ei}} = \mathbf{FL_e} + (i)$, $\mathbf{FL_{eo}} = \mathbf{FL_e} + (o)$, $\mathbf{FL_{ew}} = \mathbf{FL_e} + (i) + (o)$,
- $\mathbf{CFL_{ei}}$, $\mathbf{CFL_{eo}}$ and $\mathbf{CFL_{ew}}$, similarly,
- $\mathbf{FL_{ec}} = \mathbf{FL_e} + (Lc)$,
- $\mathbf{CFL_{ec}} = \mathbf{CFL_e} + (Lc) + (Rc)$.

Basic substructural calculi



Sequent calculus	Definition
RL	$\mathbf{FL} + (0 \Leftrightarrow 1)$
CyFL	$\mathbf{FL} + (\varphi \backslash 0 \Leftrightarrow 0 / \varphi)$
DFL	$\mathbf{FL} + (\varphi \wedge (\psi \vee \theta) \Leftrightarrow (\varphi \wedge \psi) \vee (\varphi \wedge \theta))$
$\mathbf{P_nFL}$	$\mathbf{FL} + (\varphi^n \Leftrightarrow \varphi^{n+1})$
psBL	$\mathbf{FL_w} + \{(\varphi \wedge \psi \Leftrightarrow \varphi * (\varphi \backslash \psi)), (\varphi \wedge \psi \Leftrightarrow (\psi / \varphi) * \varphi)\}$
HA	$\mathbf{FL_w} + (\varphi \Leftrightarrow \varphi^2)$
DRL	$\mathbf{RL} + (\varphi \wedge (\psi \vee \theta) \Leftrightarrow (\varphi \wedge \psi) \vee (\varphi \wedge \theta))$
IRL	$\mathbf{RL} + (\varphi \Rightarrow 1)$
CRL	$\mathbf{RL} + (\varphi * \psi \Leftrightarrow \psi * \varphi)$
GBH	$\mathbf{RL} + \{(\varphi \wedge \psi \Leftrightarrow \varphi * (\varphi \backslash \psi)), (\varphi \wedge \psi \Leftrightarrow (\psi / \varphi) * \varphi)\}$
Br	$\mathbf{RL} + (\varphi \wedge \psi \Leftrightarrow \varphi * \psi)$

Table: Some sequent calculi with their definitions.

Hard intuitionistic tautologies

Intuitionistic clique-color formulas:

$$\Theta_{n,k} := \bigwedge_{i,j}(p_{i,j} \vee q_{i,j}) \rightarrow$$

$$[(\bigwedge_{i,l}(s_{i,l} \vee s'_{i,l}) \rightarrow \alpha_n^k(\bar{p}, \bar{s}, \bar{s'})) \vee (\bigwedge_{i,l}(r_{i,l} \vee r'_{i,l}) \rightarrow \beta_n^{k+1}(\bar{q}, \bar{r}, \bar{r'}))]$$

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- $\Theta_{n,k}$ are negation-free and $\bot$-free.
- Small Roman letters denote atomic formulas
- α_n^k and β_n^{k+1} are monotone, i.e., only consist of atoms, $\wedge$ and $\vee$.
- Denote $\Theta_{n,k}$ when $k = \lfloor\sqrt{n}\rfloor$ by Θ_n.

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Theorem (Hrubeš '07, Jeřábek '09)

Formulas Θ_n are IL*-tautologies and require* IL*-Frege proofs with $2^{n^{\Omega(1)}}$ lines.*

Hard $\mathrm{FL_e}$-tautologies

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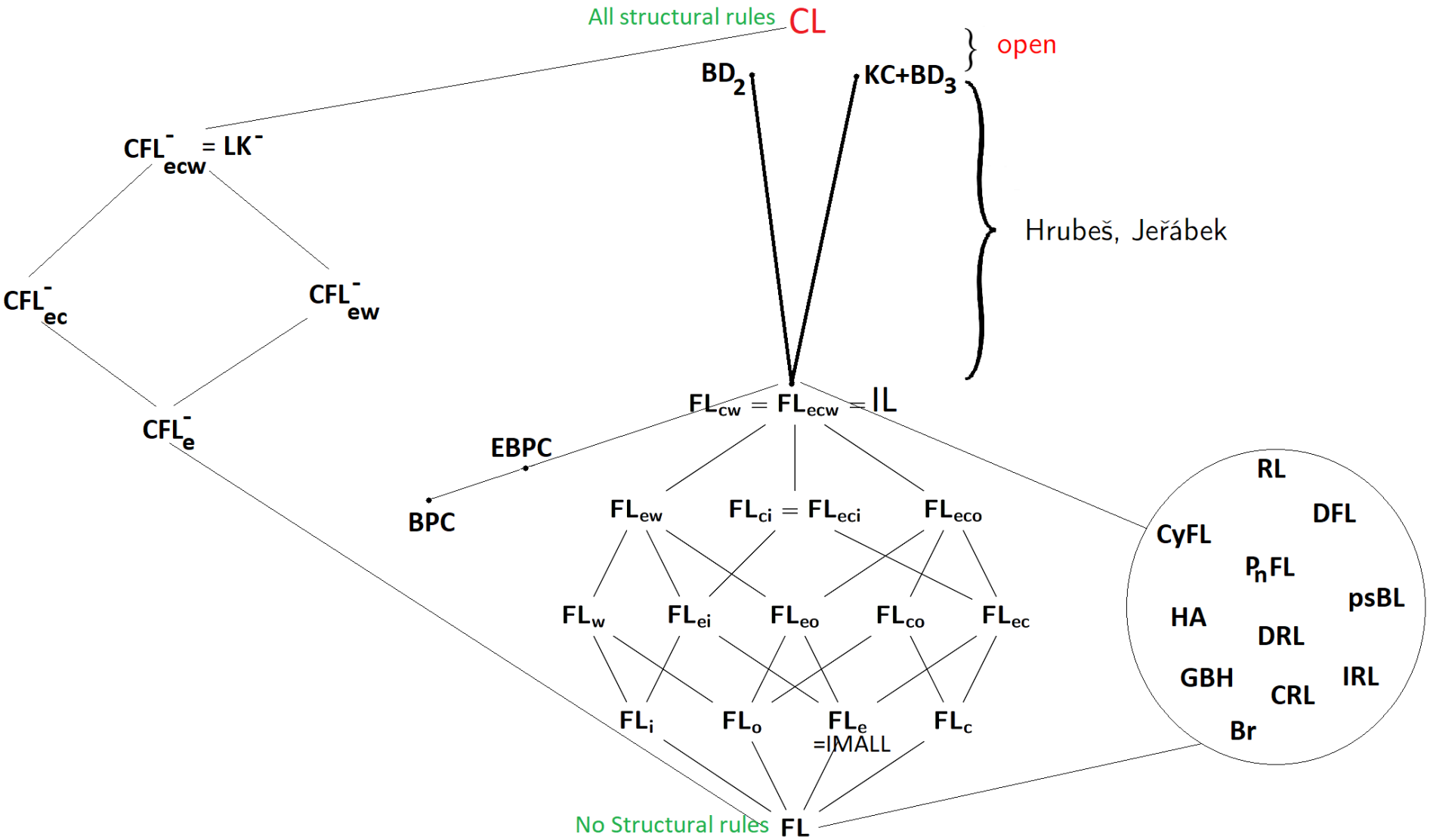
Denote $\Theta^*_{n,k}$ when $k = \lfloor \sqrt{n} \rfloor$ by Θ^*_n.

Theorem (J.)

The formulas Θ^*_n are $\mathsf{FL_e}$-tautologies. Moreover, for any substructural logic L such that $\mathsf{FL_e} \subseteq \mathsf{L} \subseteq BD_2$ or $\mathsf{FL_e} \subseteq \mathsf{L} \subseteq KC + BD_3$[a], the formulas Θ^*_n require L-Frege proofs with $2^{n^{\Omega(1)}}$ lines.

[a] $\mathsf{KC} = \mathsf{IL} + (\neg p \vee \neg\neg p)$, $\mathsf{BD_n} = \mathsf{IL} + BD_n$ where $BD_0 := \bot$ and $BD_{n+1} := p_n \vee (p_n \rightarrow BD_n)$.

Main Results at a Glance



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Theorem

The provable sequents $(\Rightarrow \Theta_n^*)$ *have* **LK**-*proofs of size* $n^{O(1)}$, *while any of their* $\mathbf{CFL_{ew}}$-*proofs*[a] *has size at least* $2^{n^{\Omega(1)}}$. *The same also holds for the Frege systems for the corresponding logics.*

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[a] Similarly, for any $\mathbf{CFL_e}$, **MALL**, $\mathbf{MALL_w}$, **LL**, or **LLW**-proof

$\Longrightarrow$ Exponential separation between $\mathbf{CFL_{ew}}$ and **LK** for $\mathrm{FL_e}$-provable formulas.

Existing Hardness Result

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The formulas $\Theta^*_{n,k}$ *are:*

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What should we do?

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Feasibly transform any **CFL$_{ew}$**-proof of Θ_n^* into an **FL$_{ew}$**-proof of Θ_n^*.
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2. For a family of formulas A (including Θ_n^*), show that $A^t \Rightarrow A$ have **feasible** **FL$_e$**-proofs.

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- Key observation: Chu's translation preserves a robust class of formulas. (2nd part of strategy)

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- Translation $(\cdot)^t$:
  - $p^t := p$ where p is an atom
  - $1^t := 1$ $\quad 0^t := \bot$ $\quad \top^t := \top$ $\quad \bot^t := \bot$
  - $(A \circ B)^t := A^t \circ B^t$ for $\circ \in \{\wedge, \vee, *\}$
  - $(A \to B)^t := (A^t \to B^t) \wedge (B^s \to A^s)$
- Translation $(\cdot)^s$:
  - $p^s := \bot$ where p is an atom
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  - $(A \wedge B)^s := A^s \vee B^s$ $\quad (A \vee B)^s := A^s \wedge B^s$
  - $(A * B)^s := (A^t \to B^s) \wedge (B^t \to A^s)$
  - $(A \to B)^s := A^t * B^s$

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For the 2nd (i.e., for a family of formulas A (including Θ_n^*), show that $A^t \Rightarrow A$ have **feasible $\mathbf{FL_e}$-proofs**), simultaneously define:

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- Conservative. $C := \bot \mid B \mid C \wedge C \mid C \vee C \mid C * C$, where B is fully conservative.

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Θ_n^* is Conservative!

The formulas

$$\alpha_n^k(\bar{p}, \bar{s}, \bar{s'}) := \bigvee_{i<n} \bigwedge_{l<k} s'_{i,l} \quad \vee \quad \bigvee_{i,j<n} \bigvee_{l<k} (s_{i,l} \wedge s_{j,l} \wedge p_{i,j}),$$

$$\beta_n^k(\bar{q}, \bar{r}, \bar{r'}) := \bigvee_{l<k} \bigwedge_{i<n} r'_{i,l} \quad \vee \quad \bigvee_{i,j<n} \bigvee_{l<m<k} (r_{i,l} \wedge r_{j,m} \wedge q_{i,j}).$$

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are fully conservative. Therefore,

$$\Theta^*_{n,k} := [\underset{i,j}{\circledast}((p_{i,j} \wedge 1) \vee (q_{i,j} \wedge 1))] \rightarrow$$

$$([\underset{i,l}{\circledast}((s_{i,l} \wedge 1) \vee (s'_{i,l} \wedge 1)) \rightarrow \alpha_n^k(\bar{p}, \bar{s}, \bar{s'})] \vee [\underset{i,l}{\circledast}((r_{i,l} \wedge 1) \vee (r'_{i,l} \wedge 1)) \rightarrow \beta_n^{k+1}(\bar{q}, \bar{r}, \bar{r'})])$$

is fully conservative and hence conservative.

Feasible Conservativity

Lemma

Let B be a fully conservative and C be a conservative formula. Then, there are $\mathbf{FL_e}$-proofs π, σ, and τ of the sequents

$$B^t \Leftrightarrow B \qquad B^s \Leftrightarrow \bot \qquad C^t \Leftrightarrow C$$

respectively, such that $\ell(\pi), \ell(\sigma) \leqslant O(|B|)$ and $\ell(\tau) \leqslant O(|C|)$.

Combining the Two Ingredients

$$\textbf{CFL}_{\textbf{ew}}\text{-proof of} \Rightarrow \Theta^*_n \xrightarrow[\text{Translation}]{\text{Chu's}} \textbf{FL}_{\textbf{ew}}\text{-proof of} \Rightarrow (\Theta^*_n)^t$$

$$\xrightarrow[\text{Lemma}]{\text{Conservativity}} \textbf{FL}_{\textbf{e}} \vdash (\Theta^*_n)^t \Leftrightarrow \Theta^*_n$$

Thus, any of $\textbf{CFL}_{\textbf{ew}}$-proof of $\Rightarrow \Theta^*_n$ has size at least $2^{n^{\Omega(1)}}$.

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PF'03
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BY ME, OF COURSE!
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* SORRY, THIS CARTOON IS TOO SMALL TO CONTAIN THE PROOF



Thank you for your attention.